

Department of Mathematics, IIT Madras
Scientific Computing Advanced Programming Lab (MA-5105)

QUIZ-II

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- This question paper has two questions, totalling to 10 marks and all are to be answered.

✓ **Q.1** Consider the QR iteration method for computing the eigenvalues of a matrix A . You are provided with the function

`qr_decomposition(A)`

which takes a matrix A as input and returns matrices Q and R such that

$$A = QR.$$

For the matrix

$$A^{(0)} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix},$$

the QR algorithm starts with $A^{(0)} = A$. At each iteration, the QR decomposition is performed on the current matrix:

$$Q^{(k)}, R^{(k)} = \text{qr_decomposition}(A^{(k)}),$$

followed by the update

$$A^{(k+1)} = R^{(k)}Q^{(k)}.$$

The iteration is continued until

$$\max_{i>j} |a_{ij}^{(k)}| < \text{tol}.$$

At convergence, the diagonal entries of the resulting matrix provide approximations to the eigenvalues.

- ✓ (a) Starting with $A^{(0)}$, write the sequence of operations required to obtain $A^{(1)}$, $A^{(2)}$, and $A^{(3)}$. Clearly indicate how `qr_decomposition()` is used at each iteration. [3 marks]

- ✓ (b) Based on the pattern observed in part (a), write **pseudocode for the complete QR algorithm**. Your pseudocode should include the initialization, iteration, stopping criterion, and extraction of the eigenvalues. [4 marks]

✓ **Q. 2** A positive integer n is given. Consider all possible **three-digit numbers** whose digits are chosen from $1, 2, \dots, n$, with repetition allowed.

For each number, calculate the product of its three digits. Write a Python program for a general value of n to determine:

- ✓ (a) The **sum of the products of the digits** of all such numbers. [2 marks]
✓ (b) The **three-digit number** whose product of digits is the largest. [1 mark]

For example, if $n = 3$, the numbers considered are

111, 112, 113, $\dots$, 333.